



Managing Digital Business (95-734)

Meeting Days, Times, Location: Mon/Wed 6-7:20PM EST Via Zoom
(reference the course schedule in the end of the syllabi)

Semester: Fall, **Year:** 2026

Units: 12, **Section(s):** Z

Instructor Information

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TA Information [If applicable]

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Course Description

The information age is continually transforming human interactions through an ever-changing landscape of trending social and economic decision making. Information technology managers, directors, and CIO / CTO's, once looked upon as the enablers of integrating and maintaining the efficiency of business technologies, are now being utilized as more of the driving force behind creating and sustaining competitive advantage over company rivals through technological innovations and implementations. Commerce has metamorphosed and is continually being transformed by these deviations to the norm, leading to new markets and potential business opportunities.

To realize success in making critical business decisions, the IT leader must have a grounded understanding of the technology in use, potential for new technology, and the ability to strategize and understand the impacts of incorporating technology in the business or industry they serve, both nationally and globally. Likewise, the IT leader must also show precise focus to the application of establishing a systematic path that encourages interactions between digital and physical markets.

In this class, we focus on two main areas: managing the business side of the entity including its people and processes through established digital business frameworks, and managing the technical roles, innovations and implications for the digital business. From a business context, the course will focus on digital business models including internet retail (eCommerce marketplaces, subscription services, curated commerce, brick & mortar establishments entering into fully autonomous commerce, how big data improves marketing potential, showrooming, etc.), requirements elicitation for understanding and establishing the needs of stakeholders both internal (employees) and external (consumers, suppliers, etc.), understanding the roles of fulfillment in the current business

environment, the difference between SCM and ERP systems to determine advantages and disadvantages of each (as well as market development impacts), and understanding the workings of B2B and P2P markets. From a technical context, this course will focus on current and emerging immersive technologies such as IoT, ePayment, IA, Blockchain, Cloud-enabled tech (Meta / Spatial Computing) & Quantum Computing.

Additionally, this course will examine IT related changes in the current marketplace through the growth of internet and mobile commerce, social networking effects on markets, Web 2.0 (and transitions into 3 and 4.0) including recommender systems and user generated /created content (UGC / UCC) and understanding the differences and nuances of digital products and services. Promoting an emphasis of the potential and proper implementation of these innovations through analysis, strategy and design, and additionally the impacts of these innovations on the organization will leverage a successful framework for maturing digital business.

Project analysis requires students to engage in both technical and managerial problem solving. The technical component of the analysis requires students to propose IT architecture for the problems highlighted in the project. The managerial component of analysis requires students to analyze the business value of the proposed solution and address the change management issues that arise in implementing any digital transformation initiative.

There are no prerequisites, but students should have taken courses that cover both information technology (Core IT Design / Development / Management / Analytics, etc.) and business/managerial topics (Financial management, business analytics, economic analysis, etc.) in prior academics. Students are highly recommended to come from a CS / IS / IT / Business background or have experience in these fields.

Learning Objectives

- Interpret, analyze, and apply technical and managerial components of emergent technologies including digital business modeling, Internet of Things, ePayment technologies, intelligent automation, blockchain technologies, cloud and quantum computing drivers.
- Identify and examine key technological challenges and rewards from systems innovations, first-mover advantages and disadvantages, and the potential disruptions or opportunities to culture, cross-industry collaborations, design thinking, and trust.
- Examine and implement current and future trends surrounding the digitization of everything across sample industries, businesses, and applicable contexts.
- Understand and adapt to the changing landscapes of economics from traditional physical to evolving digital channels.

Learning Resources

- Supplemental readings are available on Canvas & distributed in class. There is no textbook requirement.
- A “digital” reading packet containing cases and readings pertaining to lectures and assignments will be posted during the sequence of the course and may or may not contain cases requiring purchasing from HBSP or other sources.
- Feel free to utilize any favorite coding & development tools for prototypes, documentation & presentations. Ensure all on the team have the ability to access & review.

Course Web Sites: Canvas: <http://canvas.cmu.edu>

We will use Canvas for hosting initial information & announcements. If you are registered in this class, you should already have access. Your login is your Andrew ID & password. Canvas has an up-to-date copy of the syllabus, schedule, pertinent documentation, & A/V. Additionally, Canvas has links to readings & relevant sites mentioned in

our meetings with more extensive background materials.

Assessments

You will be evaluated based on your project analyses, participation in discussions, project presentations, and an individual final assessment handed out at the end of the semester.

- Your project analysis will be graded as teamwork.
- Your grades for individual class contribution & technical input will be your own.

Grades will tentatively be given at the mid-term & final. The final grade for the course is distributed among various assignments and activities posted below. Canvas should only be used to track individual grades, not cumulative. The grading system on Canvas does not accurately track the grading scheme used in this class. Please follow up with the TA and/or the instructor for grading questions. The breakdown for your work is as follows:

Student Tasks:	Total # of Tasks	Overall %
Class Contributions	variable	8%
Peer Evaluations	4	20%
Written\Presented Projects	4	52%
Final Assessment	1	20%
	9	100%

All grades utilize a check system (as follows):

GPA	Check Grade	% Grade		GPA	Check Grade	% Grade		GPA	Check Grade	% Grade
4.33	√+	98-100+		3	√-	80-89.9		0	No √	0-69.9
4	√	90-97.9		2	√--	70-79.9				

An example breakdown is as follows:

Student Name	Individual Tech	Individual Prof.	Team Tech	Team Prof.	Letter Grade
Jean Doe	√+	√	√-	√	A
Jeff Dee	√	√-	√-	√-	B
Jake Dell	√+	√+	√+	√+	A+

Class Contributions: is inherently a subjective assessment but will take the following factors into account: 1. Showing up to all required classes, paying attention with distractions removed (phone, computer, Tamagotchi, etc.) 2. Contributing to the learning in the classes via synch and asych mechanisms. 3. Contributing to asynchronous learning on canvas (discussions, surveys, etc.).

Project Analysis: You project analysis grade is based on a write-up (1 to 1.5-spacing, 12-point Time New Roman font, 1-inch margins, and no more than 20 pages total (including cover, summary, appendices, etc.) or an up to 30-minute presentation (PDF or PPT, See the course schedule for due dates & project rubric).

Peer Evaluations: Since the projects analyses and position papers are group work, I will use a peer review instrument to ensure that feedback from group members about team member contributions are considered in the determination of the final grade. 60% of the peer evaluation is dependent on your timely and well-documented submission, with the other 40% dependent on the scoring from your peers.

Final Assessment: There will be a take home final assessment handed out in the last weeks of class & is due finals week. The assessment will ask you to integrate information we have learned across the projects, required readings & lectures we've covered this semester.

Extra Credit: Can be earned by completing the [introduction profile + Exercise 0 & Individual Team Presentation Evaluations](#): Individuals on non-presenting teams are eligible to fill out a ranking survey & provide constructive feedback for team(s) presentations.

Grading Policies

Late work policy

All late deliverables are subject to a grade penalty of 10% per day past the due date/time, with a maximum of 4 days. Anything submitted beyond 4 days past the due date will receive an automatic 0. Teams & individual submissions are subject to the same policy. Any issue with meeting a deadline must be cleared up through the advisor or client / sponsor prior to the submission date / time or will be subject to the penalty.

We do not ask for health documentation to be provided directly. See the [Excuse Note Policy](#) section of the University Health Services website for more information.

Make-up work policy

Students can make up a missed assignment up until grades have been released for that assignment, subject to the late work policy. Extra credit assignments are provided during the semester to help make up missed assignments.

Re-grade policy

If you believe that your grade is inaccurate, you may request a regrade under the following conditions:

- Regrade requests must be submitted in writing within 7 days of the date when the grade was given.
- Regrade requests must outline reasons you deserve a change in your grade. Referring to another student's grade is inappropriate & irrelevant. While we do our best to apply an even standard across students, we can't discuss anyone else's grade with you, so we need to deal with the merits of your individual case only.
- We reserve the right to regrade the entire assignment requested for review & thus your grade may go up, down, or stay the same. This regrade is considered final.
- Class participation grades are inherently subjective & not subject to regrade requests. We will make notes on participation at the end of each meeting & assign end of semester grades based on these notes.

Attendance and/or participation policy

Students are expected to attend all scheduled mandatory meetings with the faculty advisors & program sponsors, as well as all presentations & final requirements. As most of this course involves team-based work including live presentations, it is imperative that students are a part of each active session including select interview sessions. Absences / lateness must be handled professionally & approved in advance.

Course Policies

Academic integrity and collaboration

Students at CMU are engaged in preparation for professional activity of the highest standards. Each profession constrains its members with both ethical responsibilities & disciplinary limits. To assure validity of the learning experience, Carnegie Mellon establishes clear standards for student work. You are required to be familiar with all university policies on this subject (see [University Policies – Academic Integrity](#)).

An extract of these policies is reproduced here:

In any presentation, creative, artistic, or research, it is the ethical responsibility of each student to produce their own academic work & identify the conceptual sources of the work submitted. Failure to do so is dishonest & is the basis for a charge of cheating or plagiarism, which is subject to disciplinary action.

Cheating includes but is not necessarily limited to:

- *Plagiarism, which is explained below.*
- *Submission of work that is not the student's own for any part of the deliverables.*
- *Submission or use of falsified data.*
- *Theft of or unauthorized access to an exam or quiz. (if applicable)*
- *Use of an alternate, stand-in or proxy during an examination. (if applicable)*

- *Use of unauthorized material including textbooks, notes or computer programs in the preparation of an assignment or during an examination. (if applicable)*
- *Supplying or communicating in any way unauthorized information to another student for work on the project outside of the team or during an examination (if applicable).*
- *Collaboration in the preparation of work on the project outside of the team. Unless specifically permitted or required by the advisor, collaboration will usually be viewed by the university as cheating. Each student is responsible for understanding the policies of the department offering any course as they refer to the amount of help & collaboration permitted in preparation of projects.*
- *Submission of the same work for credit in two projects without obtaining instructor permission.*

Plagiarism includes (but not limited to) failure to indicate source with quotation marks or footnotes where appropriate if any of the following are reproduced in the work submitted by a student:

- *A phrase, written or musical, A graphic element, A proof, Specific language, or an idea derived from the work, published or unpublished, of another person.*

One application of this plagiarism policy for this course is that you may not provide or receive information on individual completed work (applicable for work spanning more than one semester) unless approved beforehand. This includes both students from prior semesters & students from other sections in this semester. The rule of thumb is that the work provided is to be original work from the individual & the team.

Generative AI policy

I expect you to use AI (e.g., *ChatGPT and image generation tools*) in this class. In fact, some assignments may require it. Learning to use AI is an emerging skill, but the default mode of AI is to do the work for you, not with you. Learning gets punished if you use AI to create or to anchor you to its suggestions. Regardless of how you choose to use it, proper citations are required. To get started, I have provided three tutorials below (I am happy to meet and help you with these tools during office hours or after class):

- [How to use ChatGPT to boost your writing](#)
- [The practical guide to using AI to do stuff](#)
- [APA citation guidelines for GAI \(ChatGPT example\)](#)

You should also be aware of the limits of ChatGPT:

- **If you provide minimum-effort prompts, you will get low-quality results.** You will need to refine your prompts in order to get good outcomes. This will take work.
- **Don't trust everything it says.** If it gives you a number or fact, assume it's wrong unless you either know the answer or can check with another source. You will be responsible for any errors or omissions provided by the tool. It works best for topics you understand.
- **AI is a tool, but one that you need to acknowledge using.** Please include a paragraph at the end of any assignment that uses AI explaining what you used the AI for and what prompts you used to get the results. Failure to do so can constitute as an [academic integrity violation](#).
- **Be thoughtful about when this tool is useful.** Don't use it if it isn't appropriate for the case or circumstance

Resources we may use to validate work against generative AI (and that you can use) include:

- <https://www.turnitin.com/> (built into Canvas)
- <https://github.com/jwkirchenbauer/lm-watermarking>
- <https://gptzero.me/>

Some CMU strategies being employed include <https://www.cmu.edu/teaching/technology/aitools/index.html>

- You may use generative AI for the following work: **Group & individual work towards the group**
- You may NOT use generative AI to create the following work: **Surveys & individual assignments**

Mobile device policy

Devices are highly discouraged in class. Please discuss device needs with the faculty instructor.

Recording policy

Recordings of certain classes are made available on Canvas. Please check with the faculty instructor if you require recordings for classes not scheduled to be recorded.

Statements and Resources for Students

Accommodations for students with disabilities

If you have a disability and require accommodations but do not already have them approved by the Office of Disability Resources, please apply for accommodations through the Application section of the [Disability Resources Online Portal](https://rainier.accessiblelearning.com/cmu/start) (<https://rainier.accessiblelearning.com/cmu/start>). If you already have accommodations approved with Disability Resources, please use the [Accommodations Management](https://rainier.accessiblelearning.com/cmu) (<https://rainier.accessiblelearning.com/cmu>) section of the Disability Resources Online Portal to notify me about your accommodations, and discuss your accommodations and needs with me as early in the semester as possible. I will work with you to ensure that accommodation is provided as appropriate.

Inclusive excellence statement

We must treat every individual with respect. We are diverse in many ways, and recognizing this diversity is fundamental to building and maintaining an equitable and inclusive campus community. Diversity can refer to our different perspectives, lived experiences and our identities, including but not limited to race, color, national origin, sex, disability, age, sexual orientation, gender identity, pregnancy or related condition, family status, marital status, parental status, religion, ancestry, veteran status, or genetic information. What makes us different can shape the perspectives our students, faculty, and staff bring to our campus. We, at CMU, will work to promote equity and inclusion not only because they fuel excellence and innovation, but because we want to pursue justice. We acknowledge our imperfections while we also fully commit to the work, inside and outside of our classrooms, of building and sustaining a campus community that increasingly embraces our core value of inclusion.

Each of us is responsible for creating a safer, more inclusive environment.

Unfortunately, incidents of bias or discrimination do occur, whether intentional or unintentional. They contribute to creating an unwelcoming environment for individuals and groups at the university. Therefore, the university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and support, within the moment of the incident or after the incident has passed. Anyone can share these experiences using the following resources:

- **Center for Student Diversity and Inclusion:** csdi@andrew.cmu.edu, (412) 268-2150
- **Ethics Reporting Hotline:** Students, faculty, and staff can anonymously file a report by calling (844) 587-0793 or visiting cmu.ethicspoint.com.

All reports are documented and deliberated to determine if there should be any following actions. Regardless of incident type, the university will use all shared experiences to transform our campus climate to be more equitable and just.

Mandatory reporting

Carnegie Mellon University is committed to creating a learning environment where all students feel welcome, respected, and supported. Part of that commitment includes making sure that anyone who experiences discrimination or sexual misconduct has access to resources and options for support.

I want students to know their options and be able to get help when needed. Under university policy, if I'm made aware of situations involving discriminatory behavior or sexual misconduct (such as sexual harassment, sexual assault, relationship violence, or stalking), I am required to share that information with the Office for Institutional Equity and Title IX (IEX). This helps connect you with staff who can explain your options, offer resources, and support your choices moving forward.

Teaching Assistants are also mandated reporters when they learn about potential harassment, discrimination, or

sexual misconduct in their TA role.

Once a report is made, the IEX Office will reach out directly to offer information and support. You decide whether or how to respond. To learn more about IEX Office services, available resources, or reporting options, visit the [Office for Institutional Equity and Title IX website](https://www.cmu.edu/title-ix) (https://www.cmu.edu/title-ix).

If you would rather speak with someone confidentially, there are [confidential on-campus and off-campus community resources](https://www.cmu.edu/title-ix/how-to-report+-options-for-resolution/confidential-reporting.html) that are not required to report (https://www.cmu.edu/title-ix/how-to-report+-options-for-resolution/confidential-reporting.html).

Statement on student wellness

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. CMU services are available, and treatment does work. You can learn more about confidential mental health services available on campus on the [Counseling and Psychological Services \(CaPS\) website](https://www.cmu.edu/counseling) (https://www.cmu.edu/counseling). Support is always available (24/7) from CaPS: (412) 268-2922.

Food insecurity

If you are worried about affording food or feeling insecure about food, there are resources on campus that can help. Any undergraduate or graduate student can visit the CMU Pantry and receive food for free. Follow the directions on the [CMU Pantry website](https://www.cmu.edu/student-affairs/resources/cmu-pantry/) (https://www.cmu.edu/student-affairs/resources/cmu-pantry/) to schedule your visit.

Course Schedule (subject to modification):

Module	Class Topics	Launch Dates	Live Class Dates
0	Intro to the course	24-Aug	26-Aug
1a	Digital Business Modeling (DBM) 1 & Project Launch	31-Aug	
b	DBM 2	2-Sep	
c	DBM Q&A (optional)		7-Sep
d	DBM project work	9-Sep	
e	DBM Project 1:1 (optional)		14-Sep
f	DBM Presentations (mandatory)		16-Sep
2a	Internet of Everything 1 & Project Launch	21-Sep	
b	IoE 2	23-Sep	
c	IoE Q&A		28-Sep
d	IoE Project Work	30-Sep	
e	IoE Project 1:1		5-Oct
f	IoE Presentations		7-Oct
Oct 12-16 Fall Break - No Class			
3a	Intelligent Automation 1 & Project Launch	19-Oct	
b	Intelligent Automation 2	21-Oct	
c	IA Q&A		26-Oct
d	Intelligent Automation 3	28-Oct	
e	IA Project 1:1		2-Nov
f	IA Presentations		4-Nov
4a	Quantum Computing 1	9-Nov	
b	Quantum Computing 2	11-Nov	
5a	ePayment & Blockchain 1 & Project Launch	16-Nov	
b	ePayment & Blockchain 2	18-Nov	
c	eP & BC Q&A		23-Nov
d	eP & BC 3	25-Nov	
Nov26-27 Thanksgiving Break			
e	eP & BC project 1:1		30-Nov
f	eP & BC Presentations		2-Dec
6	Ind. Exam	7-Dec	
		11-Dec	

- All Q&A class meeting times **must be scheduled in advance** and are 6-7:20PM EST
- All 1:1 meetings **must be scheduled in advance** & are 20 min time slots between 6-7:20PM EST
- All class presentation times are at **6-7:20PM EST**
- All class times are **optional attendance** with exception to the presentation & guest lecture dates